

Assignment 1

Topic: Argumrnt, Modulus, Geometrical representation, Cauchy's integral formula

1. Find the modulus and argument of the following function

(i) $z = 2 + i$

(ii) $z = 1 - 3i$

(iii) $z = \frac{2-i}{3+i}(1+i)^2$

2. Describe geometrically the region of the following functions

(i) $|z - i| = 2$

(ii) $|z + 2| + |z - 2| = 3$

(iii) $\left| \frac{z-3}{z+3} \right| = 2.$

(iv) $1 < |z - 2i| \leq 2$

3. Show that $\oint_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$, where C is the circle $|z| = 5$.

4. Show that $\oint_c \frac{\sin \pi z + \cos \pi z}{(z-1)(z-2)} dz = 4\pi i$, where C is the circle $|z| = 3$.